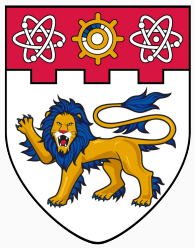
Stein's method for extreme value distributions

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Singapore - April 27, 2026

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Introduction

What is Stein's method?

- The main ingredients
- Finding a characterizing operator

An example: the Berry-Esseen bounds

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Application to Stein's method

- Stein's solution
- Rates of convergence in extreme value theory

Background: Extreme value theory

- Purpose: Estimate the frequency of extreme events + records.
  - What is the probability of occurrence of a flood strong enough to overflow a dam?
  - When? In the next decade? Century?
  - How much equity do banks need to cope with the next financial crisis? *etc.*



(a) S&P500 from 2006 to 2009



(b) 2025 floods in Viet Nam

The extreme value distributions

- *Extreme value distributions* have a fundamental role in this theory.
  - Only possible limits for maxima of i.i.d. random variables:

$$Z_n := \frac{1}{a_n} \max\left(X_1 - b_n, \ldots, X_n - b_n\right) \xrightarrow[n\to\infty]{\mathrm{d}} \mathrm{EVD}. \tag{1}$$

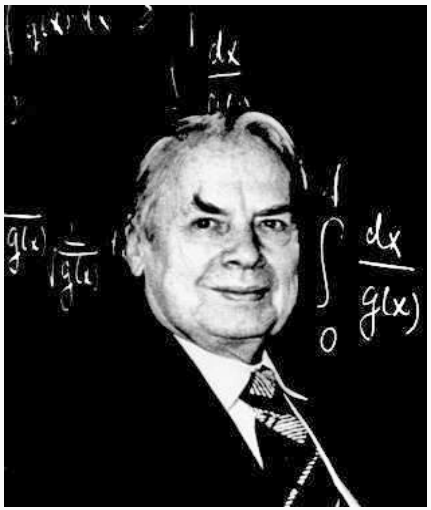
  - These distributions are necessarily *max-stable*.
  - Result known as the *Fisher-Tippett-Gnedenko theorem*.



(a) Sir R. Fisher



(b) L. H. C. Tippett



(c) B. Gnedenko

Main motivation

- Limit theorems are central in EVT: *extreme value theorem*, *Pickands-Balkema-de Haan theorem, Breiman's theorem*, *etc.*
- How much data do you need for those approximations to be good?
- Amounts to finding rates of convergence $\rightarrow$ often difficult!
- Known results rely on notions like *second-order regular variation*, *slow variation with a remainder*, *von Mises conditions*, *etc.*
- Usually yield rates of the form:

$$\mathrm{d}_K(a_n M_n + b_n, Z) = O\big(g(a_n)\big).$$

where

$$\mathrm{d}_K(X, Y) := \sup_{z \in \mathbb{R}} |F_X(z) - F_Y(z)|$$

and F_X, F_Y are the cdfs of X, Y respectively.

Stein's method

Alternative $\longrightarrow$ Stein's method!

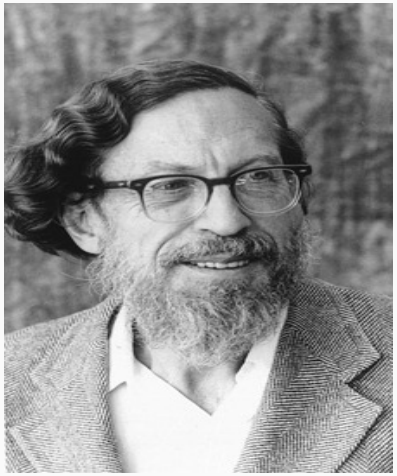


Figure 3: C. Stein

- Introduced in 1972 by Charles Stein in one of his statistics lectures!
- Framework to computes rates of convergence to a given distribution *e.g.* the normal and Poisson laws.
- Not restricted to finite dimension: [Bar88]; [Bar90]; [Dec15].

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The basic framework (1/2)

- Appropriate for *integral probability metrics*:

$$\mathrm{d}_{\mathcal{H}}(\mu,\nu) = \sup_{h\in\mathcal{H}} |\mathbb{E}_{\mu}[h] - \mathbb{E}_{\nu}[h]|.$$

- Widely used in statistics, machine learning, optimal transport, *etc.*
- **Examples:** Take $\mathcal{H}$ to be the set of
  - Indicator functions $\mathbf{1}_{(-\infty,z]} \longrightarrow$ *Kolmogorov distance* d_K
  - Indicator functions $\mathbf{1}_A \longrightarrow$ *total variation distance* d_{TV}
  - 1-Lipschitz functions $\longrightarrow$ *Wasserstein-1 distance distance* d_W
- In practice, more regularity on the test functions h may be needed *e.g.* class $\mathcal{C}^1$ 1-Lipschitz functions.

The basic framework (2/2)

- Requires two ingredients:
  1. finding a *Stein operator* $\mathscr{L}$ for the target distribution ν, *i.e.*

$$\mu = \nu \iff \mathbb{E}_\mu[\mathscr{L} g] = 0 \text{ for all } g \in \mathcal{G},$$

     for some large enough function space $\mathcal{G}$;
  2. finding $g_h \in \mathcal{G}$ solving *Stein's equation*:

$$\mathscr{L} g_h = h - \mathbb{E}_\nu[h], \ h \in \mathcal{H}. \tag{2}$$

- Then

$$\mathbb{E}_\mu[h] - \mathbb{E}_\nu[h] = \mathbb{E}_\mu\left[\mathscr{L} g_h\right]$$

  and so

$$\mathrm{d}_{\mathcal{H}}(\mu, \nu) = \sup_{h \in \mathcal{H}} |\mathbb{E}_\mu[\mathscr{L} g_h]| \leq \sup_{g \in \mathcal{G}} |\mathbb{E}_\mu[\mathscr{L} g]|.$$

- In practice, one can take $\mathcal{G} := \{g_h, \ h \in \mathcal{H}\}$.

How to construct a characterizing operator $\mathscr{L}$ for a given ν?

Via the density approach [Ste+04]

- Assume ν has a differentiable density $p : (a, b) \to \mathbb{R}_+^*$.
- Let $\psi(x) := (\log p(x))'$ be the *score function*.
- Then for $Z \sim \nu$:

$$\mathbb{E}\big[g'(Z) + \psi(Z)g(Z) - p(b)g(b) + p(a)g(a)\big] = 0. \tag{3}$$

- Can be adapted to the discrete case:
  - **Example:** Set $p_n = \frac{\lambda^n}{n!} e^{-\lambda}$, $n \in \mathbf{N}$. Then $np_n = \lambda p_{n-1}$, hence

$$\mathbb{E}\big[Zg(Z)\big] = \lambda \mathbb{E}\big[g(Z+1)\big],$$

    where $Z \sim \mathcal{P}(\lambda)$.

Examples

- $\nu = \mathcal{N}(0,1) \quad \longrightarrow \quad p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, hence:

$$\mathbb{E}[g'(Z)] = \mathbb{E}\big[Zg(Z)\big].$$

  Thus we can take $\mathscr{L}g(x) = -xg(x) + g'(x)$.

- $\nu = \mathcal{VP}(\alpha) \quad \longrightarrow \quad p(x) = \alpha x^{-(\alpha+1)}$, hence:

$$\mathbb{E}\big[g'(Z)\big] = (\alpha+1)\mathbb{E}\big[Z^{-1}g(Z)\big] - \alpha g(1).$$

- $\nu = \mathrm{B}(\alpha,\beta) \quad \longrightarrow \quad p(x) = \frac{1}{Z_{\alpha,\beta}} x^{\alpha-1}(1-x)^{\beta-1}$, hence:

$$\mathbb{E}\big[Z(1-Z)g_0'(Z)\big] = \mathbb{E}\big[\big(\beta Z - \alpha(1-Z)\big)g_0(Z)\big]$$

  where $g_0(x) := x(1-x)g(x)$.

Via the generator approach [Bar88]

- Find a Markov process $(X_t)_{t\geq 0}$ whose semi-group $(P_t)_{t\geq 0}$

$$P_t h(x) := \mathbb{E}\big[h(X_t) \,|\, X_0 = x\big]$$

quantizes ν:

$$\mathbb{E}_\nu[P_t h] = \mathbb{E}_\nu[h].$$

- Then $\mathscr{L}$ is a Stein operator for ν and $\mathcal{H}$, with

$$\mathscr{L} := \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} P_t.$$

- Besides, $\mathscr{L}$ is invertible on $\mathcal{H}_0$ and $g_h = \mathscr{L}^{-1} h^*$ equals:

$$g_h = -\int_0^\infty P_t h \,\mathrm{d}t, \tag{4}$$

and where

$$\mathcal{H}_0 := \{h \in \mathcal{H},\ \mathbb{E}_\nu[h] = 0\}.$$

Examples (1/2)

- The *Laguerre process* [AS15]:

$$\mathrm{d}Z_{t,x} = (a - \lambda Z_{t,x})\,\mathrm{d}t + \sqrt{2Z_{t,x}}\,\mathrm{d}B_t$$

whose stationary distribution is $\nu = \Gamma(\alpha, \lambda)$ and generator is:

$$\mathscr{L}f(x) = (\alpha - \lambda x)f'(x) + xf''(x).$$

- The *overdamped Langevin diffusion* [GM15]

$$\mathrm{d}\boldsymbol{Z}_{t,\boldsymbol{x}} = \frac{1}{2}(\nabla \log p)(\boldsymbol{Z}_{t,\boldsymbol{x}})\,\mathrm{d}t + \mathrm{d}\boldsymbol{B}_t,$$

where p is the density function of the target distribution ν.

Examples (2/2)

- The *standard OU semi-group* [Bar88]:

$$P_t h(x) = \mathbb{E}\left[h\left(e^{-t}x + \sqrt{1-e^{-2t}}Z\right)\right]$$

with $Z \sim \mathcal{N}(0,1)$, has generator:

$$\mathscr{L} g(x) = -xg'(x) + g''(x).$$

- The α*-stable OU semi-group* [AH23]

$$P_t h(\boldsymbol{x}) := \mathbb{E}\left[h\left(e^{-t}\boldsymbol{x} + (1-e^{-\alpha t})^{\frac{1}{\alpha}}\boldsymbol{Z}\right)\right]$$

where $\phi_{\boldsymbol{Z}}(\boldsymbol{x}) = e^{-\frac{\|\boldsymbol{x}\|^\alpha}{2}}$, has generator:

$$\mathscr{L} g(x) = -\langle \boldsymbol{x}, \nabla g(\boldsymbol{x})\rangle + \int_{\mathbb{R}^d} \left\langle \nabla g(\boldsymbol{x}+\boldsymbol{u}) - \nabla g(\boldsymbol{x})\right), \boldsymbol{u}\rangle \, \mathrm{d}\mu_\alpha(u).$$

and μ_α is an α-homogeneous Lévy measure.

Why does it work?

- The *stability postulate* is the key argument:

$$e^{-t}\boldsymbol{Z} + (1 - e^{-\alpha t})^{\frac{1}{\alpha}}\boldsymbol{Z}' \stackrel{\mathrm{d}}{=} \boldsymbol{Z}.$$

  if $\boldsymbol{Z}, \boldsymbol{Z}'$ are i.i.d. with characteristic function $e^{-\frac{\|\boldsymbol{x}\|^{\alpha}}{2}}$, for $\alpha \in (0, 2]$.
- Gives the identity:

$$P_t \circ P_s = P_{t+s}.$$

- A general notion of stability can be defined through the concept of *convex cone* [DMZ08].

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Statement of the result

Theorem (The Berry-Esseen bounds [NP12])

Let $X_1, \ldots, X_n$ be centered i.i.d. rv with variance $\sigma^2 = 1$ and finite third moment $\mathbb{E}\left[|X_1|^3\right]$. Then:

$$\mathrm{d}_K(Z_n, Z) \leq C\mathbb{E}[|X_1|^3]\frac{1}{\sqrt{n}}$$

for some universal constant $C > 0$ and where

$$Z_n := \frac{1}{\sqrt{n}} S_n := \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i.$$

Step 1: Solve Stein's equation

- For every $z \in \mathbb{R}$, set

$$\varphi(z) := \exp(-z^2/2)/\sqrt{2\pi} \quad \text{and} \quad \Phi(z) := \int_{-\infty}^{z} \varphi(y)\, \mathrm{d}y.$$

- Set $h_z(x) := \mathbf{1}_{(-\infty,z]}(x)$ and g_z an associated Stein solution:

$$-xg_z(x) + g_z'(x) = \mathbf{1}_{(-\infty,z]}(x) - \Phi(z). \tag{5}$$

- The only bounded solution is

$$g_z(x) = \frac{1}{\varphi(x)} \Phi\big(\min(x,z)\big)\Phi\big(-\max(x,z)\big)$$

- One has the estimates:

$$\|g_z\|_\infty \leq \frac{\sqrt{2\pi}}{4} \quad \text{and} \quad \|g_z'\|_\infty \leq 1.$$

Step 2: Bound $\mathbb{E}[\mathscr{L} g_z(Z_n)]$ (1/2)

- Prove that $\mathbb{E}[\mathscr{L} g_z(Z_n)] \simeq 0$ *i.e.*

$$\mathbb{E}\big[Z_n g_z(Z_n)\big] \simeq \mathbb{E}\big[g_z'(Z_n)\big].$$

- Set $Z_n^i := \sum_{j \neq i} X_i/\sqrt{n}$.
- Start from one side and use Taylor's formula at Z_n^i:

$$\begin{aligned}
\mathbb{E}\big[Z_n g_z(Z_n)\big] &= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathbb{E}\big[X_i g_z(Z_n)\big] \\
&= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathbb{E}\big[X_i g_z(Z_n^i + X_i/\sqrt{n})\big] \\
&= \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\big[X_i^2 g_z'(Z_n^i + \theta X_i/\sqrt{n})\big]
\end{aligned}$$

where $\theta \sim \mathcal{U}_{[0,1]}$.

Step 2: Bound $\mathbb{E}[\mathscr{L} g_z(Z_n)]$ (2/2)

- Recall that $\mathbb{E}[X_i^2] = 1$, so that

$$\mathbb{E}\big[g_z'(Z_n^i)\big] = \mathbb{E}\big[X_i^2 g_z'(Z_n^i)\big].$$

- Therefore

$$\begin{aligned}\mathbb{E}\big[Z_n g_z(Z_n)\big] - \mathbb{E}\big[g_z'(Z_n)\big] \\ = \mathbb{E}\big[X_i^2 g_z'(Z_n^i + \theta X_i/\sqrt{n})\big] - \mathbb{E}\big[X_i^2 g_z'(Z_n^i + X_i/\sqrt{n})\big]\end{aligned}$$

- The error is thus bounded by:

$$\frac{1}{n}\sum_{i=1}^{n} \sup_{z\in\mathbb{R}} \mathbb{E}\Big[X_i^2 \big|g_z'(Z_n^i + \theta X_i/\sqrt{n}) - g_z'(Z_n^i)\big|\Big].$$

- It can be bounded accurately thanks to the properties of Stein's solution g_z.

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Max-stability

Definition (Multivariate max-stable distribution)

Let $\alpha \in \mathbb{R}$ and define for any $r \in \mathbb{R}_+$ and $\boldsymbol{x} \in \mathbb{R}^d$:

$$\mathrm{d}_r^\alpha \boldsymbol{x} := \begin{cases} r^{\frac{1}{\alpha}} \boldsymbol{x} & \text{if } \alpha \neq 0 \\ \boldsymbol{x} + \log r \mathbf{1} & \text{if } \alpha = 0. \end{cases}$$

A distribution ν on $\mathbb{R}^d$ is *max-stable with index* α if for any i.i.d. $\boldsymbol{Z}, \boldsymbol{Z}'$ with law ν and $a, b \geq 0$, there exists some $c \geq 0$ s.t.:

$$\mathrm{d}_a^\alpha \boldsymbol{Z} \oplus \mathrm{d}_b^\alpha \boldsymbol{Z}' \overset{\mathrm{d}}{=} \mathrm{d}_c^\alpha \boldsymbol{Z},$$

where $\boldsymbol{x} \oplus \boldsymbol{y} := \big(\max(x_1, y_1), \ldots, \max(x_d, y_d)\big)$.

Notice that d^α satisfies:

$$\mathrm{d}_a^\alpha \circ \mathrm{d}_b^\alpha = \mathrm{d}_{ab}^\alpha \text{ and } \mathrm{d}_r^\alpha(\boldsymbol{x} \oplus \boldsymbol{y}) = \mathrm{d}_r^\alpha \boldsymbol{x} \oplus \mathrm{d}_r^\alpha \boldsymbol{y}.$$

Example of extreme value distributions

- **Examples:** (1D)
  - The *Weibull distribution* $\mathcal{W}(\alpha)$ on $\mathbb{R}_-$:

$$F(x) = e^{-(-x)^{-\alpha}} \longrightarrow aZ \oplus bZ' \stackrel{\mathrm{d}}{=} (a^\alpha + b^\alpha)^{\frac{1}{\alpha}} Z.$$

  - The *Gumbel distribution* $\mathcal{G}(0,1)$ on $\mathbb{R}$:

$$F(x) = e^{-e^{-x}} \longrightarrow (Z + \log a) \oplus (Z' + \log b) \stackrel{\mathrm{d}}{=} Z + \log(a+b).$$

  - The *Fréchet distribution* $\mathcal{F}(\alpha)$ on $\mathbb{R}_+$:

$$F(x) = e^{-\frac{1}{x^\alpha}} \longrightarrow aZ \oplus bZ' \stackrel{\mathrm{d}}{=} (a^\alpha + b^\alpha)^{\frac{1}{\alpha}} Z.$$

- **Examples:** (2D)
  - Set $\boldsymbol{Z} := (Z_1, Z_2)$, with $Z_1, Z_2 \sim \mathcal{F}(\alpha)$ and i.i.d. Then

$$a\boldsymbol{Z} \oplus b\boldsymbol{Z}' \stackrel{\mathrm{d}}{=} (a^\alpha + b^\alpha)^{\frac{1}{\alpha}} \boldsymbol{Z}.$$

  - Same for $\boldsymbol{Z} := (Z_1, Z_1)$.

The extreme value theorem

- The previous 1D examples are the only possible non-degenerate limits for a maximum of n i.i.d. random variables $(X_i)_{i\geq 1}$:

$$Z_n := \frac{1}{a_n}\bigoplus_{i=1}^{n}(X_i - b_n) \xrightarrow[n\to\infty]{\mathrm{d}} \begin{cases} \mathcal{W}(\alpha) \\ \mathcal{G}(0,1) \\ \mathcal{F}(\alpha). \end{cases} \tag{6}$$

- Max-stable counterpart of the *generalized central limit theorem*.

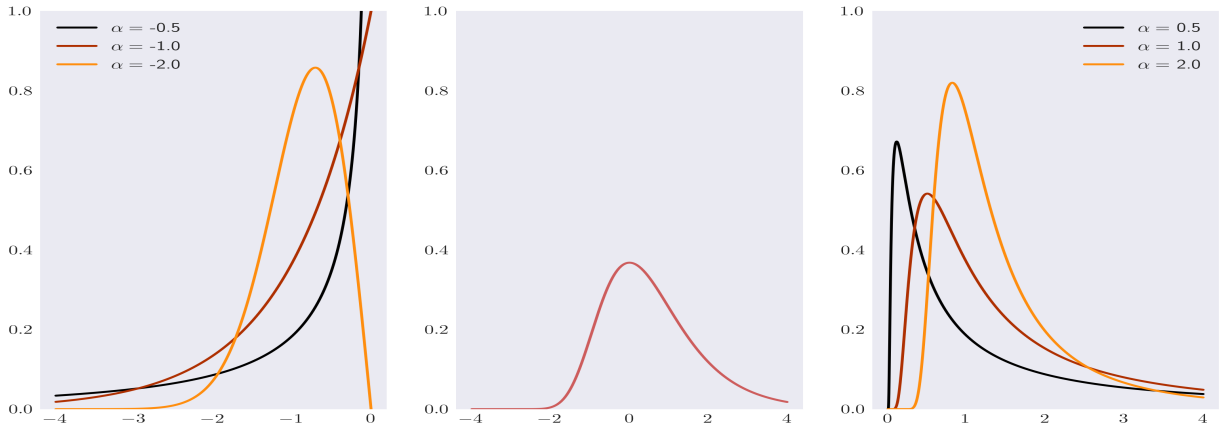


Figure 4: P.d.f of $\mathcal{W}(-\alpha)$, $\mathcal{G}(0,1)$ and $\mathcal{F}(\alpha)$ for $\alpha \in \{0.5, 1, 2\}$

The polar decomposition

Definition (α-homogeneous measure)

A measure μ is said to be an *α-homogeneous measure* on a subset E of $\overline{\mathbb{R}}^d$ if

$$\mu(\mathrm{d}_r^\alpha B) = r^{-1}\mu(B).$$

for every $B \in \mathscr{B}(E)$ and $r > 0$.

- For $\alpha > 0$, let μ be an α-homogeneous measure on $E_0 := [\mathbf{0}, +\infty]^d \setminus \{\mathbf{0}\}$, *i.e.*

$$\mu\big(r^{\frac{1}{\alpha}} B\big) = r^{-1}\mu(B).$$

- Fix a reference norm $\|\cdot\|$ and set

$$\begin{array}{rccc} T & : & E_0 & \to & \mathbb{R}_+^* \times \mathbb{S}_+^{d-1} \\ & & \boldsymbol{x} & \mapsto & \big(\|\boldsymbol{x}\|, \frac{\boldsymbol{x}}{\|\boldsymbol{x}\|}\big) \end{array}$$

where $\mathbb{S}_+^{d-1} := \{\boldsymbol{u} \in \mathbb{R}_+^d,\ \|\boldsymbol{u}\| = 1\}$.

The de Haan-LePage representation

- Set $\rho_\alpha[r, +\infty] := r^{-\alpha}$.
- Then we have the *polar decomposition*:

$$T^*\mu = \rho_\alpha \otimes \sigma$$

for some finite measure σ on $\mathbb{S}_+^{d-1}$.

- Any max-stable random vector $\boldsymbol{Z}$ can be represented as:

$$\boldsymbol{Z} \overset{\mathrm{d}}{=} \bigoplus_{\boldsymbol{x} \in \eta_1} \boldsymbol{x} \overset{\mathrm{d}}{=} \bigoplus_{(r,\boldsymbol{u}) \in \eta_2} r\boldsymbol{u}, \tag{7}$$

where η_1 is a PPP on $E_{\mathbf{0}}$ whose intensity measure is some α-homogeneous μ, and η_2 a marked PPP on $\overline{\mathbb{R}}_+^* \times \mathbb{S}_+^{d-1}$ with intensity measure $\rho_\alpha \otimes \sigma$.

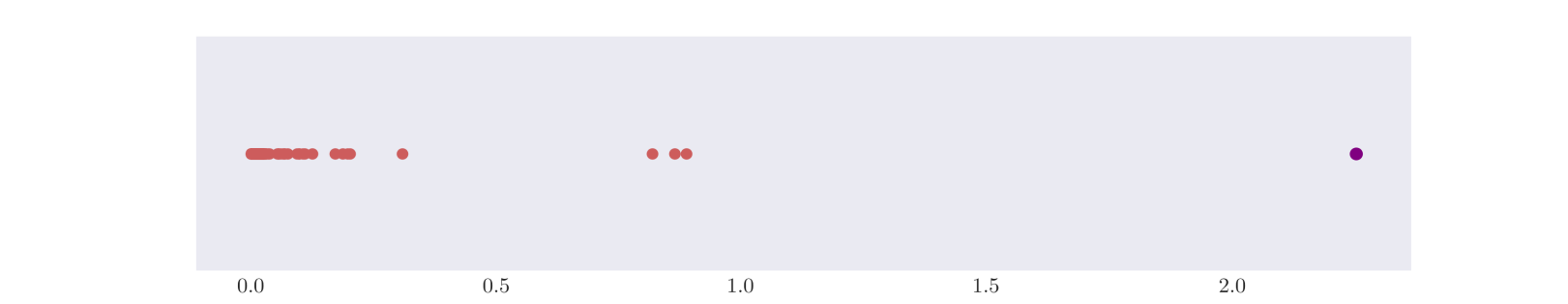
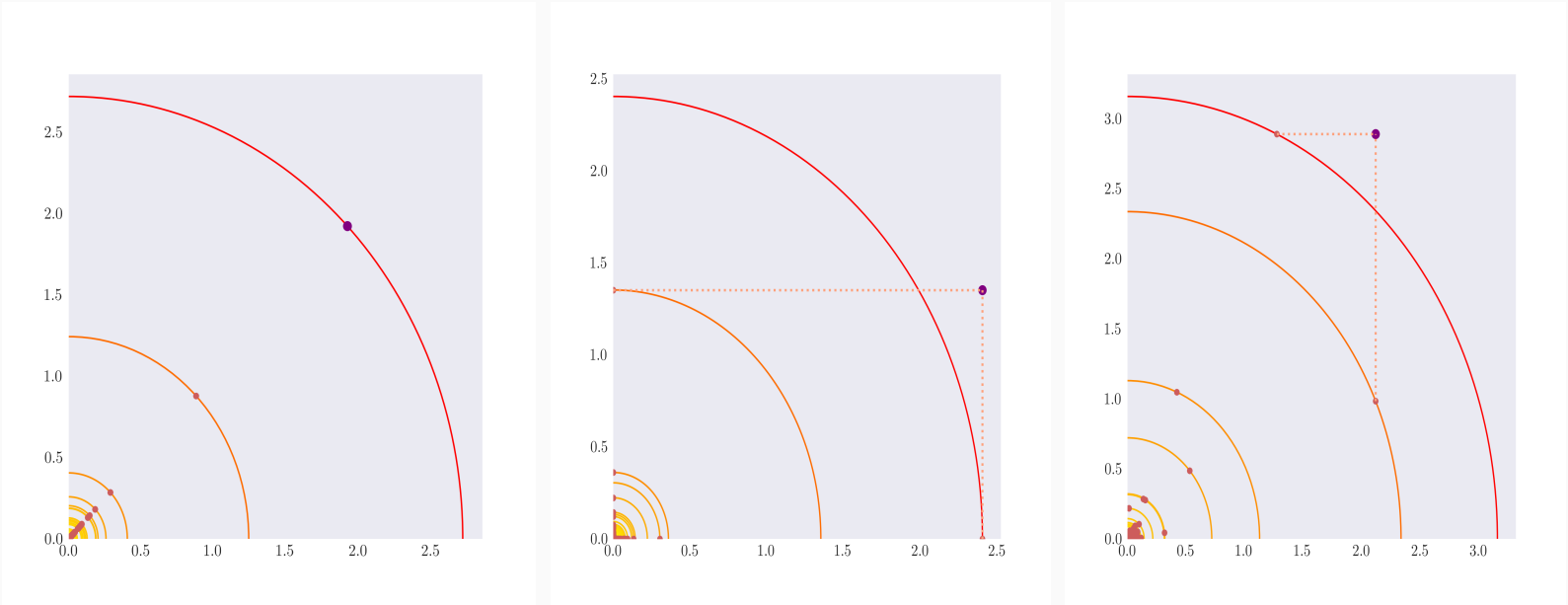


Figure 5: PPP on $\mathbb{R}_+$ with intensity measure $\mathrm{d}\rho_1(r) = \frac{1}{r^2}\,\mathrm{d}r$



(a) Deterministic **(b)** Bernoulli **(c)** Uniform

Figure 6: Three PPP with different choices of angular measure σ

Parametrization of max-stable random vectors

- μ is called the *exponent measure* of $\boldsymbol{Z}$.
- σ is called the *angular measure* of $\boldsymbol{Z}$.
- We will note $\boldsymbol{Z} \sim \mathcal{MS}(\alpha, \sigma)$ or sometimes $\boldsymbol{Z} \sim \mathcal{MS}(\mu)$.
- The c.d.f. of $\boldsymbol{Z}$ is:

$$\mathbb{P}(\boldsymbol{Z} \leq \boldsymbol{x}) = e^{-\mu[\mathbf{0}, \boldsymbol{x}]^c}.$$

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The max-stable semi-groups

Definition ([CD25b])

Define the *Fréchet semi-group* with index $\alpha > 0$ by

$$\begin{aligned}\mathbf{P}_t^{\alpha,\sigma} h(\boldsymbol{x}) &:= \mathbb{E}\Big[h\big(\mathrm{d}^{\alpha}_{e^{-t}}\boldsymbol{x} \oplus \mathrm{d}^{\alpha}_{(1-e^{-t})}\boldsymbol{Z}\big)\Big] \\ &= \mathbb{E}\Big[h\big(e^{-\frac{t}{\alpha}}\boldsymbol{x} \oplus (1-e^{-t})^{\frac{1}{\alpha}}\boldsymbol{Z}\big)\Big],\ \boldsymbol{x} \in \mathbb{R}^d_+,\end{aligned}$$

where $\boldsymbol{Z} \sim \mathcal{MS}(\alpha, \sigma)$ and $t \geq 0$.

- Define T_α by

$$T_\alpha h(\boldsymbol{x}) = \begin{cases} h(\boldsymbol{x}^\alpha) & \text{if } \alpha \neq 0 \\ h(\log \boldsymbol{x}) & \text{if } \alpha = 0. \end{cases}$$

- Then

$$\mathbf{P}_t^{\alpha,\sigma} = T_\alpha \mathbf{P}_t^{1,\sigma} T_\alpha^{-1}. \tag{8}$$

Generator

Theorem (Limit behaviour)

The family $(\mathbf{P}_t^{1,\sigma})_{t\geq 0}$ is an ergodic semi-group which admits the distribution $\mathcal{MS}(1,\sigma)$ as its stationary measure.

Proposition (Generator)

The generator $\mathscr{L}_{1,\sigma}$ of $(\mathbf{P}_t^{1,\sigma})_{t\geq 0}$ is:

$$\begin{aligned}\mathscr{L}_{1,\sigma} f(\boldsymbol{x}) &= -\langle \boldsymbol{x}, \nabla f(\boldsymbol{x})\rangle + \int_{E_0} \big(f(\boldsymbol{x}\oplus \boldsymbol{y}) - f(\boldsymbol{x})\big)\, \mathrm{d}\mu(\boldsymbol{y}) \\ &= -\langle \boldsymbol{x}, \nabla f(\boldsymbol{x})\rangle + \int_{E_{\mathrm{pol}}} \big(f(\boldsymbol{x}\oplus r\boldsymbol{u}) - f(\boldsymbol{x})\big)\, \frac{1}{r^2}\, \mathrm{d}r\, \mathrm{d}\sigma(\boldsymbol{u}),\end{aligned}$$

where $E_{\mathrm{pol}} = \mathbb{R}_+^* \times \mathbb{S}_+^{d-1}$.

Commutation rules

Theorem (Semi-group commutation rule)

Set

$$\mathbf{D}_{1,\sigma} f(\boldsymbol{x}) := \int_{E_{\mathrm{pol}}} \big(f(\boldsymbol{x} \oplus r\boldsymbol{u}) - f(\boldsymbol{x})\big) \frac{1}{r^2} \, \mathrm{d}r \, \mathrm{d}\sigma(\boldsymbol{u}).$$

Then

$$\mathbf{D}_{1,\sigma} \mathbf{P}_t^{1,\sigma} = e^{-t} \mathbf{P}_t^{1,\sigma} \mathbf{D}_{1,\sigma}.$$

Proposition (Infinitesimal commutation rule)

$$[\mathscr{L}_{1,\sigma}, \mathbf{D}_{1,\sigma}] := \mathscr{L}_{1,\sigma} \circ \mathbf{D}_{1,\sigma} - \mathbf{D}_{1,\sigma} \circ \mathscr{L}_{1,\sigma} = \mathbf{D}_{1,\sigma}.$$

Poincaré inequality

- Set $\mathcal{E}_{1,\sigma}(f) := -\mathbb{E}\big[f(\boldsymbol{Z})\mathscr{L}_{1,\sigma}f(\boldsymbol{Z})\big]$.
- The *energy form* $\mathcal{E}_{1,\sigma}$ of $(\mathbf{P}_t^{1,\sigma})_{t\geq 0}$ equals

$$\mathcal{E}_{1,\sigma}(f) = \int_{\mathbb{R}_+^d} \mathbb{E}\Big[\Big(f(\boldsymbol{Z}\oplus r\boldsymbol{u}) - f(\boldsymbol{Z})\Big)^2\Big]\frac{1}{r^2}\,\mathrm{d}r\,\mathrm{d}\sigma(\boldsymbol{u})$$

Proposition (Poincaré inequality)

For every nice function f, the following holds true

$$\mathbb{V}\big(f(\boldsymbol{Z})\big) \leq 2\mathcal{E}_{1,\sigma}(f). \tag{9}$$

Thus :

$$\|\mathbf{P}_t^{1,\sigma}f - \mathbb{E}[f(\boldsymbol{Z})]\|_{\mathbf{L}^2} \leq e^{-\frac{t}{2}}\|f - \mathbb{E}[f(\boldsymbol{Z})]\|_{\mathbf{L}^2}.$$

Representation of $(\mathbf{P}_t^\alpha)_{t\geq 0}$ as a Markov process

- There exists a Markov process $(X_t)_{t\geq 0}$ s.t.

$$\mathbf{P}_t^\alpha h(x) = \mathbb{E}\big[h(X_t) \,|\, X_0 = x\big].$$

- That process can be expressed in terms of *extremal integrals* (see [ST05]):

$$X_t := e^{-\frac{t}{\alpha}} X_0 \oplus \overset{e}{\int_0^t} e^{-\frac{1}{\alpha}(t-s)} \,\mathrm{d}M_\alpha(s),$$

where M_α is a α-Fréchet random sup-measure with Lebesgue control measure.

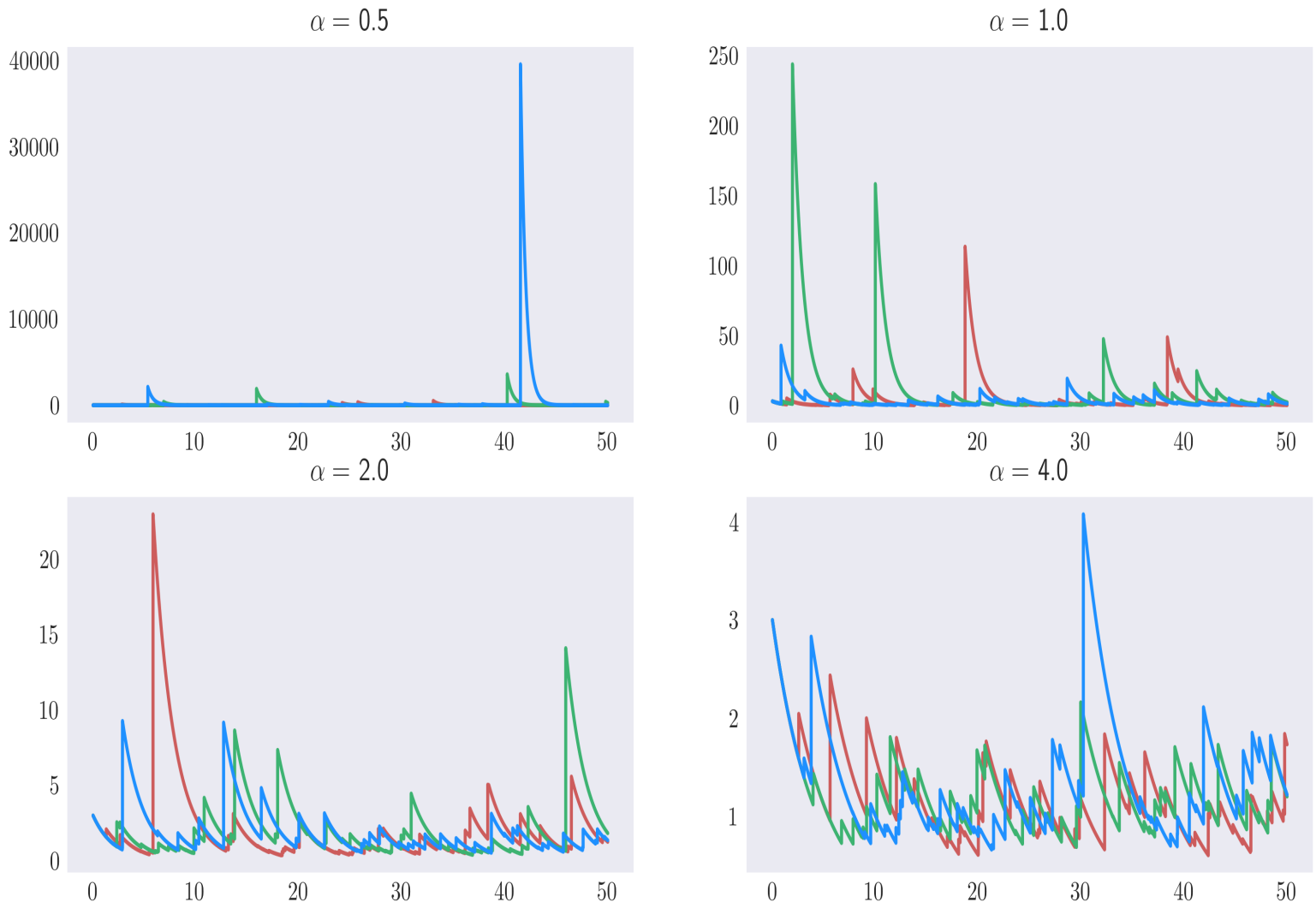


Figure 7: Simulations of X_t starting at $X_0 = 3$ with $\alpha \in \left\{\frac{1}{2}, 1, 2, 4\right\}$

Plan

Introduction

What is Stein's method?

- The main ingredients
- Finding a characterizing operator

An example: the Berry-Esseen bounds

A crash course in extreme value theory

The smart path method for extreme value distributions

Application to Stein's method

- Stein's solution
- Rates of convergence in extreme value theory

Stein equation

Proposition (Characterization of max-stable distributions)

Let $\boldsymbol{X}$ be a random vector. Then $\boldsymbol{X} \sim \mathcal{MS}(\alpha, \sigma)$ iff

$$\mathbb{E}\big[\langle \boldsymbol{X}, \nabla g(\boldsymbol{X}) \rangle\big] = \alpha \mathbb{E}\big[\mathbf{D}_{\alpha,\sigma} g(\boldsymbol{X})\big],$$

for all nice functions $g : E_{\mathbf{0}} \to \mathbb{R}$.

- Let h be a test function and denote by g_h the solution to:

$$h(\boldsymbol{x}) - \mathbb{E}[h(\boldsymbol{Z})] = \mathscr{L}_{\alpha,\sigma} h(\boldsymbol{x}), \ \boldsymbol{x} \in E_{\mathbf{0}}. \tag{10}$$

- What are the properties of g_h? Regularity? Boundedness?

Properties of g_z in 1D (1/2)

In the Kolmogorov case, the Stein solution g_z satisfies:

Proposition (Regularity)

Let $z \in \mathbb{R}_+^*$.

1. g_z is non-decreasing and constant over $[0, z]$:

$$g_z(x) = -\int_0^\infty \left(e^{-\frac{1-e^{-t}}{z^\alpha}} - e^{-\frac{1}{z^\alpha}} \right) \mathrm{d}t, \ x \in [0, z].$$

2. g_z is differentiable over $\mathbb{R}_+^* \setminus \{z\}$:

$$g_z'(x) = \begin{cases} \frac{\alpha}{x} e^{-\left(\frac{1}{z^\alpha} - \frac{1}{x^\alpha}\right)} & \text{if } x > z \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

Properties of g_z in 1D (2/2)

Proposition (Asymptotic behaviour)

3 We have the equivalent

$$g_z(x) \underset{x \to +\infty}{\sim} \alpha e^{-\frac{1}{z^\alpha}} \log x.$$

4 g_z' satisfies the inequality

$$|g_z'(x)| \leq \frac{\alpha}{x} \mathbf{1}_{\{x>z\}}, \ x > 0. \tag{12}$$

5 We have:

$$\mathbf{D}_\alpha g_z(x) = e^{-\frac{1}{z^\alpha}} \left(e^{\min(\frac{1}{x^\alpha}, \frac{1}{z^\alpha})} - 1 \right), \ x > 0. \tag{13}$$

Consequently, $\sup_{z \in \mathbb{R}_+} \|\mathbf{D}_\alpha g_z\|_\infty \leq 1$.

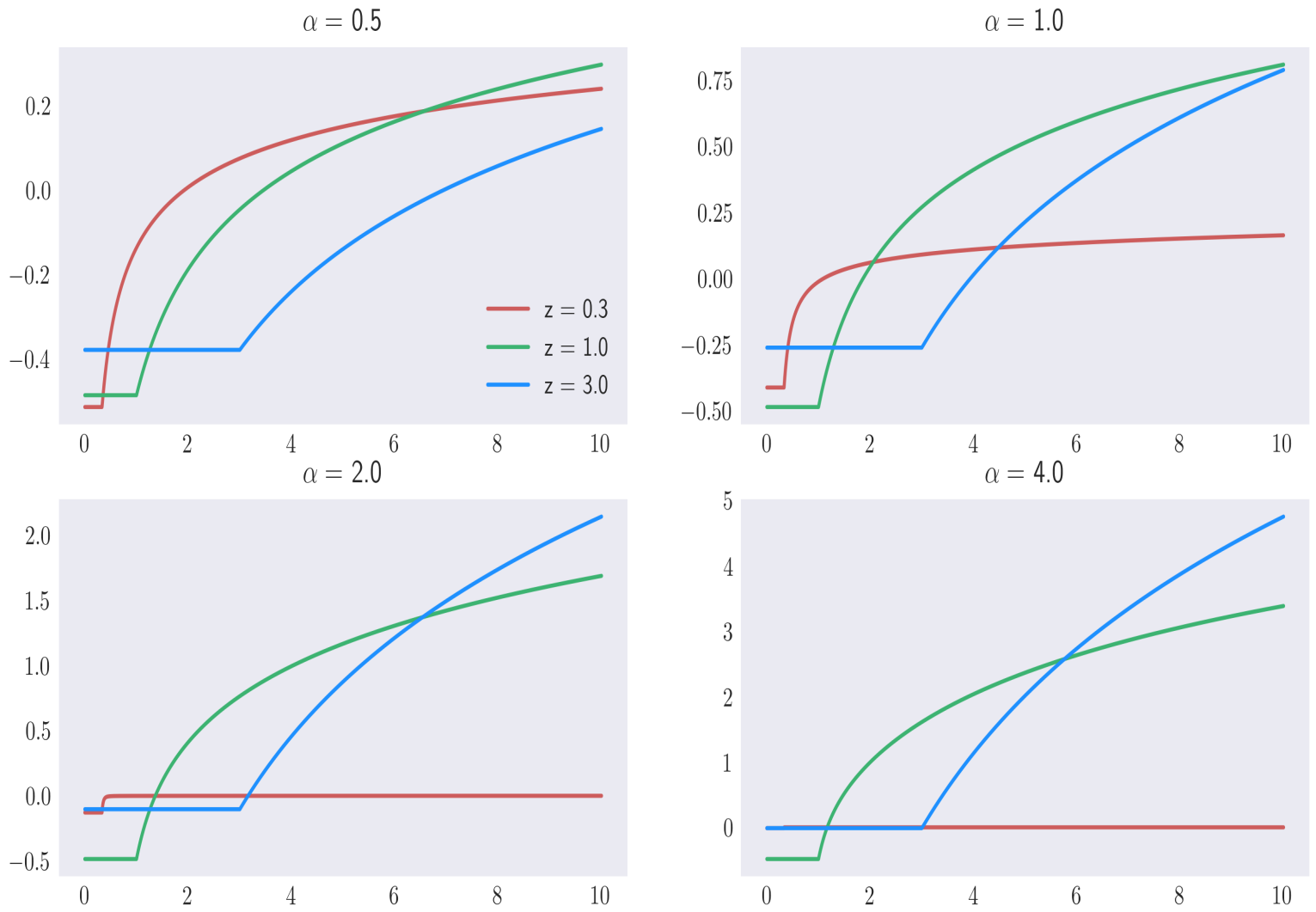


Figure 8: Graphs of g_z over $[1, 10]$ for $z \in \left\{\frac{1}{3}, 1, 3\right\}$ and $\alpha \in \left\{\frac{1}{2}, 1, 2, 4\right\}$

Main assumptions

- **Notations:** Let $X_1, \ldots, X_n$ be i.i.d. random variables with c.d.f. F_X.
- Assume there are $\alpha > 0$ and $\rho < 0$ s.t.

$$\overline{F}_X := 1 - F_X \in 2\mathrm{RV}_{\alpha,\rho}.$$

Definition (Second-order regular variation)

Let $\alpha \in \mathbb{R}$ and $\rho \leq 0$. A function $G : \mathbb{R} \to \mathbb{R}_+^*$ is $2\mathrm{RV}_{\alpha,\rho}$ if $\exists A : \mathbb{R}_+^* \to \mathbb{R}$ vanishing at infinity s.t.:

$$\lim_{t\to\infty} \frac{1}{A(t)} \left(\frac{G(tx)}{G(t)} - x^\alpha \right) = x^\alpha \left(\frac{x^\rho - 1}{\rho} \right), \quad x > 0. \tag{14}$$

If $\rho = 0$, the right-hand side is equal to $x^\alpha \log x$.

Goal

- Set also $a_n := F^{\leftarrow}(1 - 1/n)$ and

$$Z_n := a_n^{-1} \max(X_1, \ldots, X_n).$$

- Then

$$Z_n \underset{n\to\infty}{\longrightarrow} \mathcal{F}(\alpha).$$

- **Goal:** Bound the distance between Z_n and $\mathcal{F}(\alpha)$.

Theorem (WIP)

Under the previous assumptions, there exists some $C_X > 0$ s.t.

$$\mathrm{d}_K(Z_n, Z) \leq C_X \left(n^{-1} + n^{\alpha/\rho}\right).$$

Beyond the EVT

- Stein's method is not restricted to i.i.d. random variables!
- Can be used to get a rate of convergence for the *coupon collector problem* in a Wasserstein-like distance [CD25a]:

$$Z_n = \sum_{i=1}^{n} \frac{T_i - \log n}{n}$$

where T_i is the time of apparition of the i-th new coupon.

- And also to find a rate of convergence for the *de Haan-LePage series* [CD25c] when taking only the first n points of the PPP:

$$Z_n := \bigoplus_{i=1}^{n} r_i \boldsymbol{u}_i,$$

with $\big((r_i, \boldsymbol{u}_i)\big)_i$ a PPP with intensity measure $\rho_\alpha \otimes \sigma$.

Bibliography i

[AH23] B. Arras and C. Houdré. **"On some operators associated with non-degenerate symmetric α-stable probability measures".** In: *Potential Analysis* 59.4 (2023), pp. 1643–1694.

[AS15] B. Arras and Y. Swan. **"A stroll along the gamma".** In: *Stochastic Processes and their Applications* 127 (Nov. 2015). DOI: 10.1016/j.spa.2017.03.012.

[Bar88] A. D. Barbour. **"Stein's method and Poisson process convergence".** In: *Journal of Applied Probability* 25.A (1988), 175–184. DOI: 10.2307/3214155.

[Bar90] A. D Barbour. **"Stein's method for diffusion approximations".** In: *Probability theory and related fields* 84.3 (1990), pp. 297–322.

Bibliography ii

[CD25a] B. Costacèque and L. Decreusefond. **“Convergence rate for the coupon collector’s problem with Stein’s method”.** In: *Stochastic Processes and their Applications* (2025), p. 104835.

[CD25b] B. Costacèque and L. Decreusefond. **“Functional analysis of multivariate max-stable distributions”.** In: *arXiv preprint arXiv:2509.02200* (2025).

[CD25c] B. Costacèque and L. Decreusefond. **“Stein’s method for max-stable random vectors”.** In: *arXiv preprint arXiv:2507.00463* (2025).

[Dec15] L. Decreusefond. **“The Stein-Dirichlet-Malliavin Method”.** In: *ESAIM: Proceedings* (2015), p. 11. DOI: 10.1051/proc/201551003. URL: https://partage.imt.fr/index.php/s/FGmZiZMLXz3FJfN.

Bibliography iii

[DMZ08] Y. Davydov, I. Molchanov, and S. Zuyev. **"Strictly stable distributions on convex cones".** In: *Electronic Journal of Probability* 13 (2008), no. 11, 259–321. DOI: 10.1214/EJP.v13-_487. URL: https://doi.org/10.1214/EJP.v13-_487.

[GM15] J. Gorham and L. Mackey. **"Measuring sample quality with Stein's method".** In: *Advances in neural information processing systems* 28 (2015).

[NP12] I. Nourdin and G. Peccati. ***Normal approximations with Malliavin calculus. From Stein's method to universality*.** 1st ed. CUP, 2012.

Bibliography iv

[ST05] S.A. Stoev and M.S. Taqqu. **“Extremal stochastic integrals: a parallel between max-stable processes and alpha-stable processes”.** In: *Extremes* 8 (2005), pp. 237–266.

[Ste+04] C. Stein et al. **“Use of exchangeable pairs in the analysis of simulations”.** In: *Lecture Notes-Monograph Series* (2004), pp. 1–26.